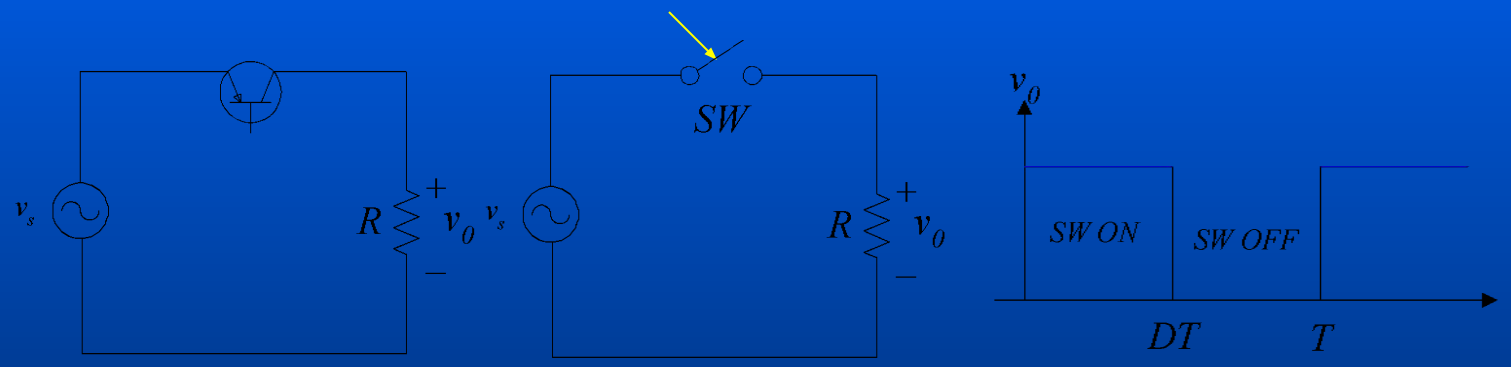
# 제 6 장 직류변환장치

## *DC to DC Converter*
## *DC→ DC*

# 6.1 개요

- AC의 전력변환: 변압기을 사용 쉽게 가능함
- DC의 전력변환: 변압기로는 어려움
- DC-DC Converter를 사용하면 가능
- 종류
  - 강압 초퍼(Buck chopper)
  - 승압 초퍼(Boost chopper)
  - 강압 및 승압 초퍼(Buck & Boost chopper)

# 6.2 Basic Switching Converter



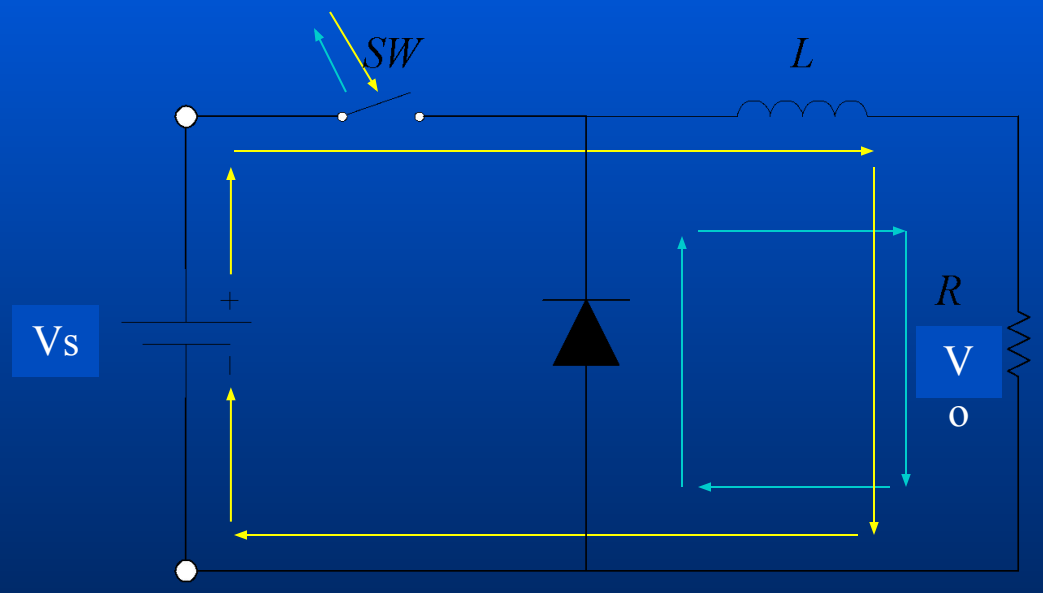

- Switching 방식에 의한 DC 제어

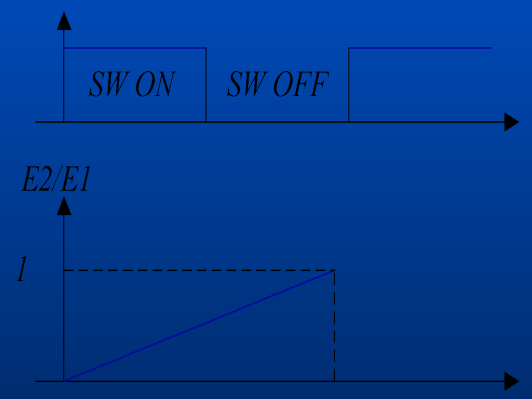
$$V_0 = \frac{1}{T}\int_0^T v_0(t)dt = \frac{1}{T}\int_0^{DT} V_S(t)dt$$

$$= \frac{V_S}{T}\int_0^{DT} 1dt = \frac{V_S}{T}\left[t\right]_0^{DT} = \frac{V_S}{T}(DT-0) = V_S D$$

- Duty Cycle(D)=Switch On 시간(DT)/주기(T)

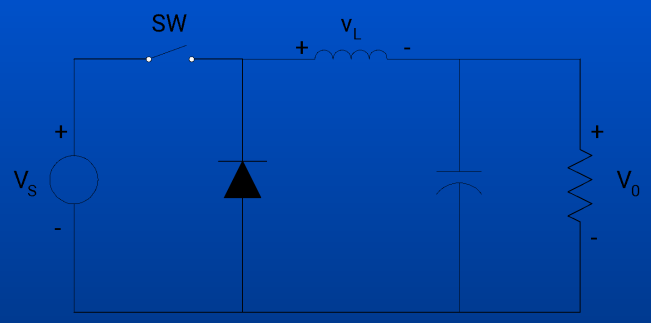
# 6.3 강압: Buck Converter(Step-down)

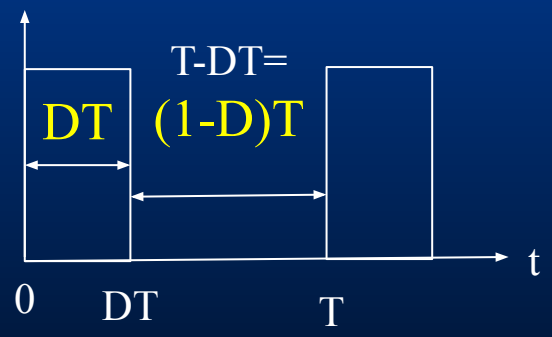



- 입력(Vs) 보다 출력(Vo) 이 작음

# 6.3 Buck Converter: 전압




- Switch On

$$v_L = V_S - V_0 = L\frac{di_L}{dt} = L\frac{\Delta i_L}{\Delta t} = L\frac{\Delta i_L}{DT}$$

$$\therefore \Delta i_L = \frac{1}{L}(V_S - V_0)DT$$

- Switch Off

$$v_L = -V_0 = L\frac{di_L}{dt} = L\frac{\Delta i_L}{\Delta t} = L\frac{\Delta i_L}{(1-D)T}$$

$$\therefore \Delta i_L = -\frac{1}{L}V_0(1-D)T$$

- 충전전류+방전전류=0

$$\cancel{\frac{1}{L}}(V_S - V_0)D\cancel{T} \times \frac{L}{T} - \cancel{\frac{1}{L}}V_0(1-D)\cancel{T} \times \frac{L}{T} = 0 \quad \times \frac{L}{T}$$

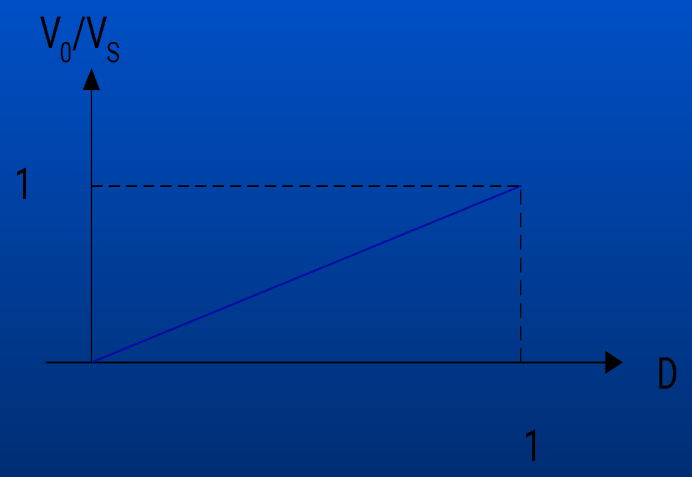
$$V_S D - \cancel{V_0 D} - V_0 + \cancel{V_0 D} = 0 \qquad \therefore V_0 = V_S D$$

# 6.3 Buck Converter: 출력

- 충전전류+방전전류=0

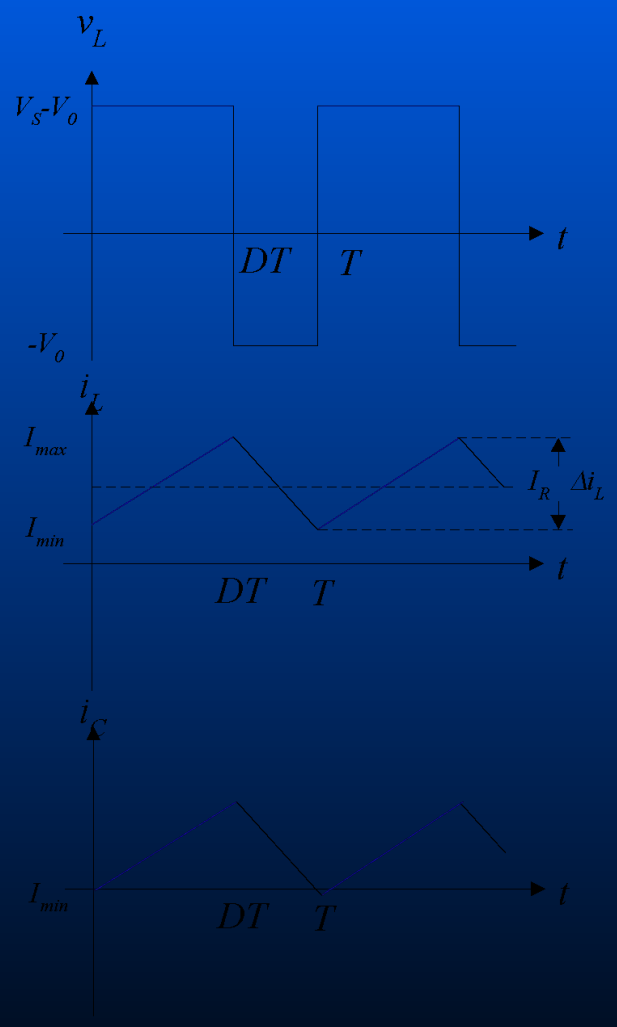
$$D \le 1, \quad V_0 = V_S D, \quad \therefore \frac{V_0}{V_S} = D$$

| D | 출력/입력($V_0$ /$V_S$ ) |
|---|---|
| 0 | 0 |
| 0.2 | 0.2 |
| 0.5 | 0.5 |
| 0.8 | 0.8 |
| 1.0 | 1.0 |




- 강압(Step-down)
- 입력 대비 출력이 낮음

# 6.3 Buck Converter: 리플



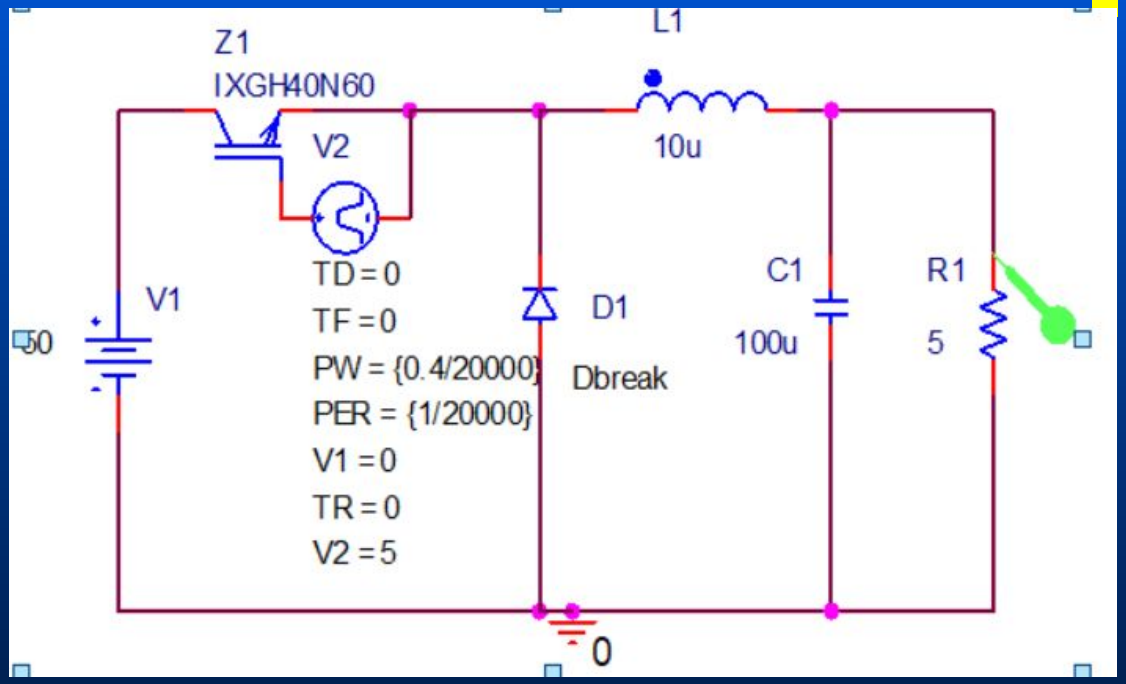

- Ripple

$$v_L = -V_0 = L\frac{di_L}{dt}$$

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = -\frac{1}{L}V_0$$

$$\Delta i_L = (1-D)T\left(-\frac{1}{L}V_0\right)$$

# 예제 6-1 벅 변환기



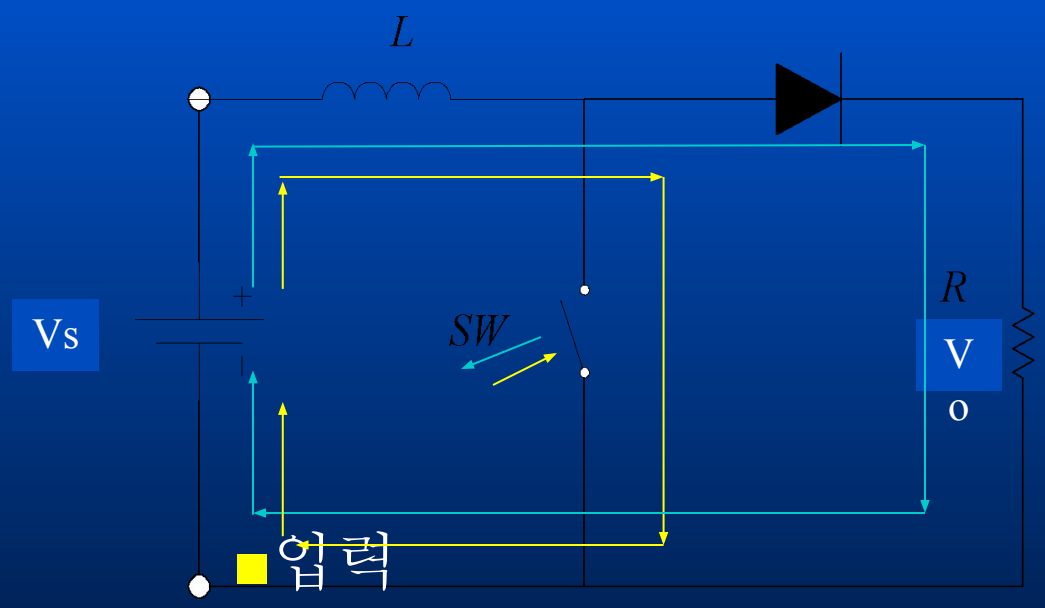

## *Parameter*

- ✔ $V_S$=50 V
- ✔ L=10 μH
- ✔ C=100 μF
- ✔ f=20 kHz(T=50 μs)
- ✔ D=0.4(T=20 μs)
- ✔ R=20 ohm
- ✔ Step
  =0 1 ms

## *Find*

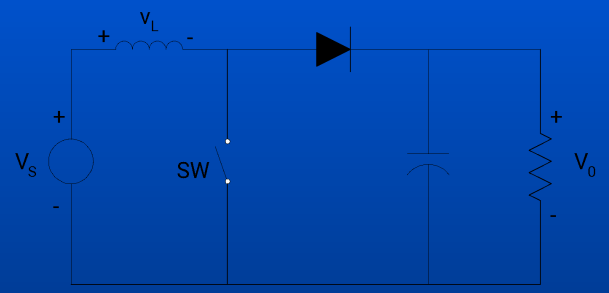
- ✔ 출력전압
- ✔ 전류 최대, 최소값

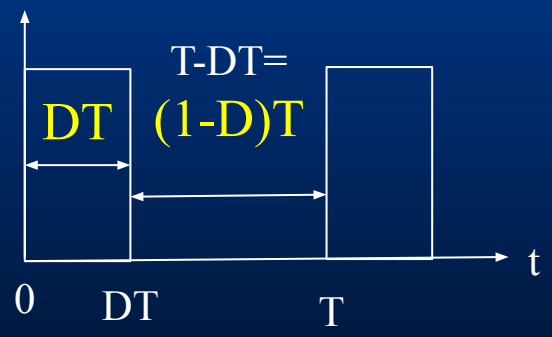
# 6.5 승압: Boost Converter(Step-up)



- 입력(Vs)보다 출력(Vo) 이 커짐

# 6.3 Boost Converter: 전압




- Switch On

$$v_L = V_S = L\frac{di_L}{dt} = L\frac{\Delta i_L}{\Delta t} = L\frac{\Delta i_L}{DT}$$

$$\therefore \Delta i_L = \frac{1}{L}V_S DT$$

- Switch Off

$$v_L = V_S - V_0 = L\frac{di_L}{dt} = L\frac{\Delta i_L}{\Delta t} = L\frac{\Delta i_L}{(1-D)T}$$

$$\therefore \Delta i_L = \frac{1}{L}(V_S - V_0)(1-D)T$$

- 충전전류+방전전류=0

$$\cancel{\frac{1}{L}}V_S D\cancel{T} \times \frac{L}{T} + \cancel{\frac{1}{L}}(V_S - V_0)(1-D)\cancel{T} \times \frac{L}{T} = 0 \times \frac{L}{T}$$

$$\cancel{V_S D} + V_S - \cancel{V_S D} - V_0 + V_0 D = 0$$

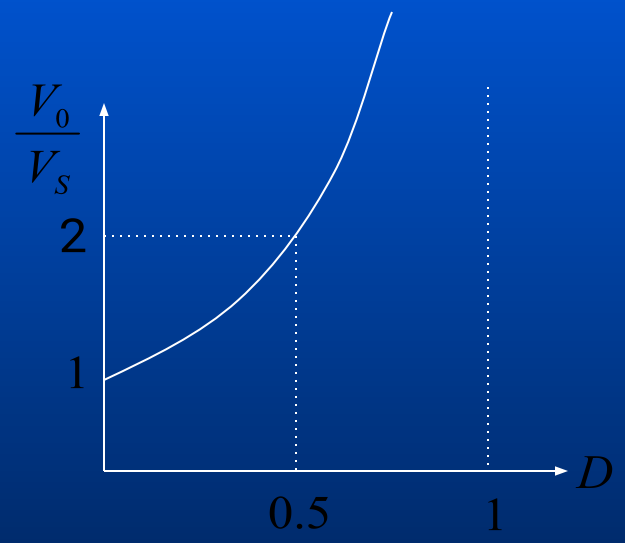
$$\therefore V_0 = \frac{V_S}{(1-D)}$$

# 6.5 Boost Converter: 출력

- 충전전류+방전전류=0

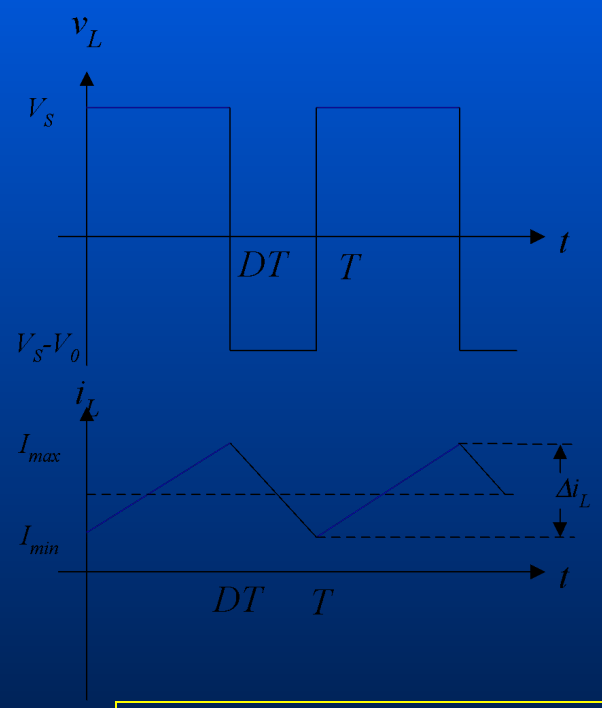
$$D \le 1,\ V_0 = \frac{V_S}{(1-D)},\ \therefore \frac{V_0}{V_S} = \frac{1}{(1-D)}$$

| D | 출력/입력 | $(V_0/V_S)$ |
|---|---|---|
| 0 | 1/(1−0) | 1 |
| 0.2 | 1/(1−0.2) | 1.25 |
| 0.5 | 1/(1−0.5) | 2.0 |
| 0.8 | 1/(1−0.8) | 5.0 |
| 1.0 | 1/(1−1) | ∞ |




- 승압(Step-up)
- 입력 대비 출력이 높음

# 6.5 Boost Converter: 인덕터전류




- 인덕터의 평균전류

$$V_S I_L = \frac{V_0^2}{R} = \frac{\left(\frac{V_0}{(1-D)}\right)^2}{R} = \frac{V_S^2}{(1-D)^2 R}$$

$$\therefore I_L = \frac{V_S}{(1-D)^2 R}$$

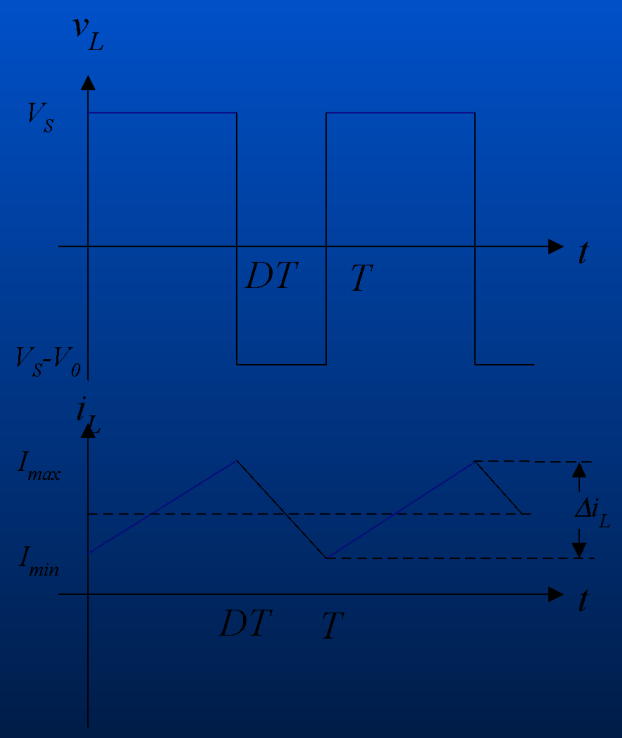
- 인덕터의 전류 최소, 최대값

$$\left( \frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_S}{L} \right)$$

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_S}{(1-D)^2 R} + \frac{V_S DT}{2L}$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_S}{(1-D)^2 R} - \frac{V_S DT}{2L}$$

# 6.5 Boost Converter: 스위칭 주파수




- 전류의 연속조건:
  최소값이 0 보다 커야함

$$I_{\min} = 0 = \frac{V_S}{(1-D)^2 R} - \frac{V_S DT}{2L}$$

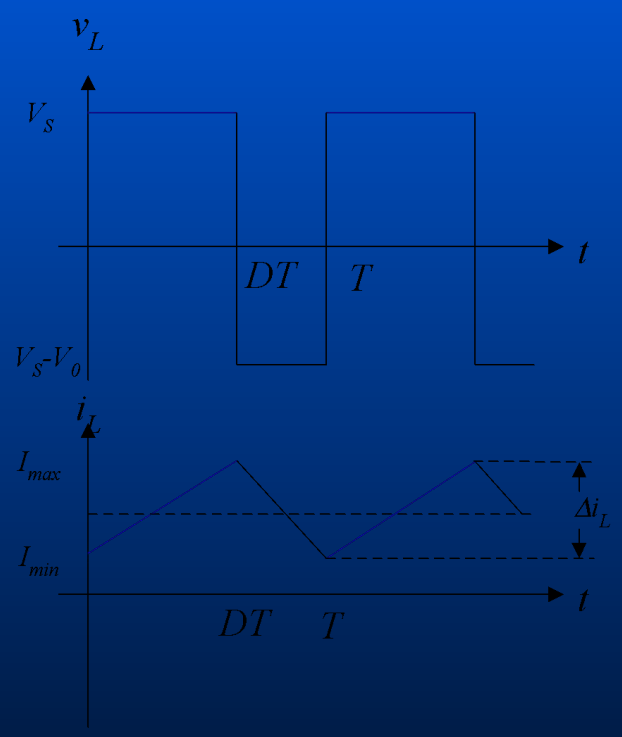
$$\frac{V_S}{(1-D)^2 R} = \frac{V_S DT}{2L} = \frac{V_S D}{2Lf}$$

- 인덕터 및 스위치의 조합

$$(Lf)_{\min} = \frac{D(1-D)^2 R}{2}$$

$$\therefore L_{\min} = \frac{D(1-D)^2 R}{2f}$$

# 6.5 Boost Converter: 전압의 맥동




- 커패시터 전하 변화량

$$|\Delta Q| = \left(\frac{V_0}{R}\right) DT = C\Delta V_0$$

- 맥동

$$\Delta V_0 = \frac{V_0 DT}{RC} = \frac{V_0 D}{RCf}$$

$$\frac{\Delta V_0}{V_0} = \frac{DT}{RC} = \frac{D}{RCf}$$

# 예제 6-3 부스트 변환기의 설계

- 12V→30V로 승압, 맥동율: 1% 미만, R=50ohm
- Duty 비

$$\frac{V_0}{V_S} = \frac{1}{(1-D)}, \quad D = 1 - \frac{V_0}{V_S} = 1 - \frac{12}{30}$$

- 인덕터 최소값: f=25kHz로 함(가청주파수 보다 높게)

$$L_{\min} = \frac{D(1-D)^2 R}{2f} = \frac{0.6(1-0.6)^2 50}{2(25000)} = 96\mu H$$

# 예제 6-3 부스트 변환기의 설계(2)

- 인덕터 전류

$$I_L = \frac{V_S}{(1-D)^2 R} = \frac{12}{(1-0.6)^2 50} = 1.5\,A$$

$$\frac{\Delta i_L}{2} = \frac{V_S D}{2Lf} = \frac{(12)(0.6)}{2(120\times10^{-6})(25000)} = 1.2\,A$$

$$I_{\max} = 1.5 + 1.2 = 2.7\,A$$

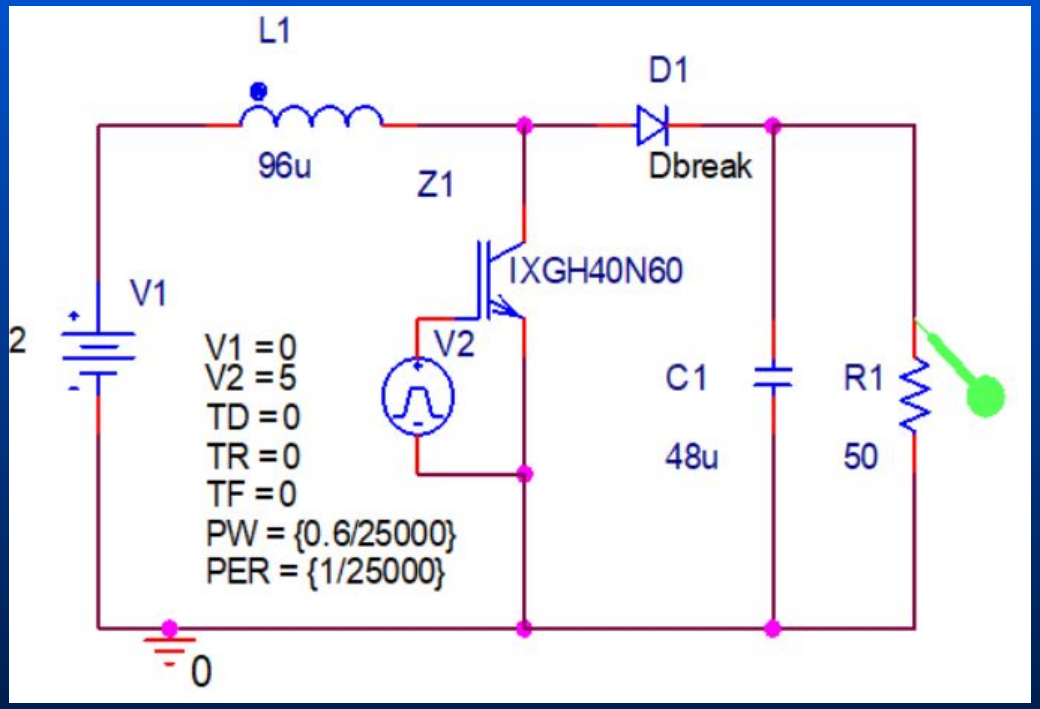
$$I_{\min} = 1.5 - 1.2 = 0.3\,A$$

- 인덕터 전류

$$\frac{\Delta V_0}{V_0} < 1\%$$

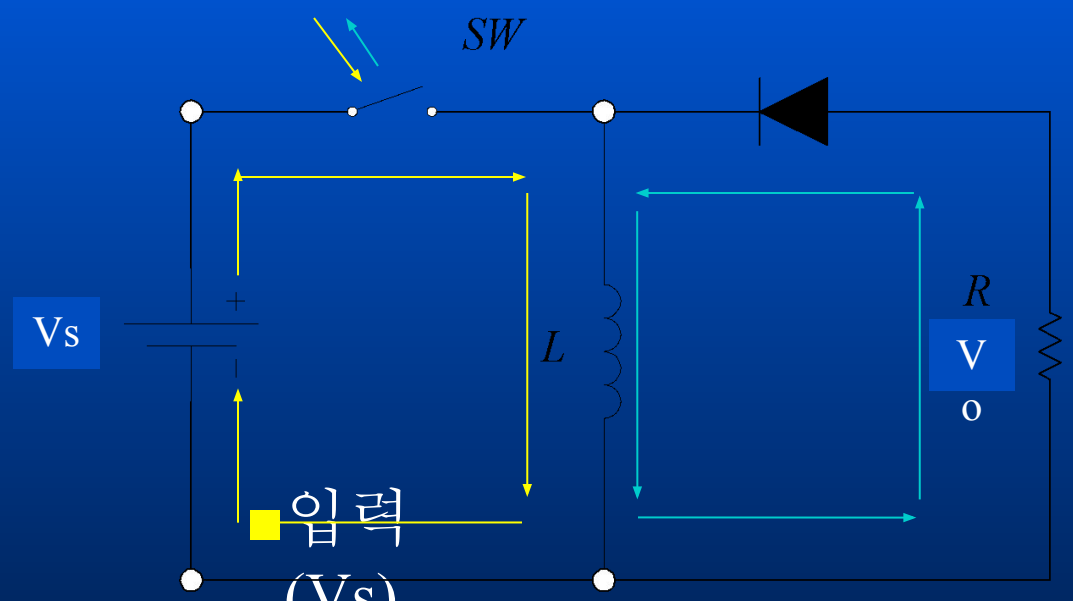
$$C >= \frac{D}{Rf(\Delta V_0 / V_0)} = \frac{0.6}{(50)(25\times10^3)(0.01)} = 48\mu F$$

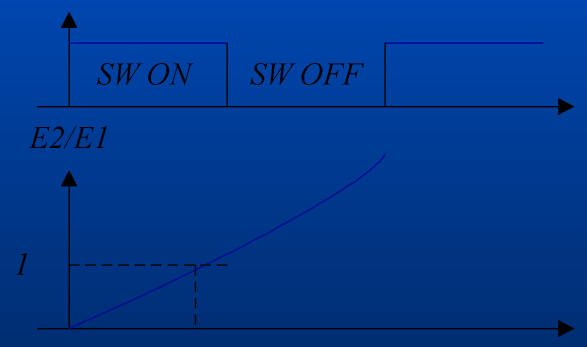
# 예제 6-3 부스트 변환기(Pspice 해)




- *Parameter*
  - ✔ $V_S$=12 V
  - ✔ L=96 μH
  - ✔ C=48 μF
  - ✔ f=25 kHz(T=40 μS)
  - ✔ D=0.6 (D=24 μS)
  - ✔ R=50 ohm
  - ✔ Step
    =0 1 ms
- *Find*
  - ✔ 출력전압
  - ✔ 전류 최대, 최소값

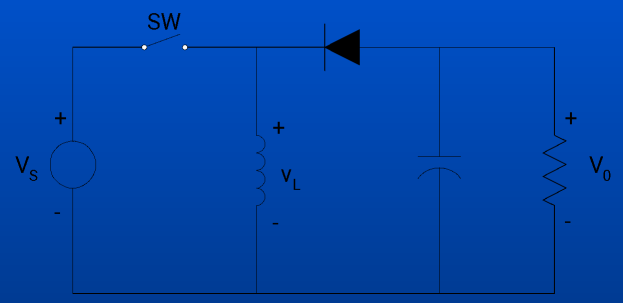
# 6.6 강압 및 승압: Buck-Boost Converter

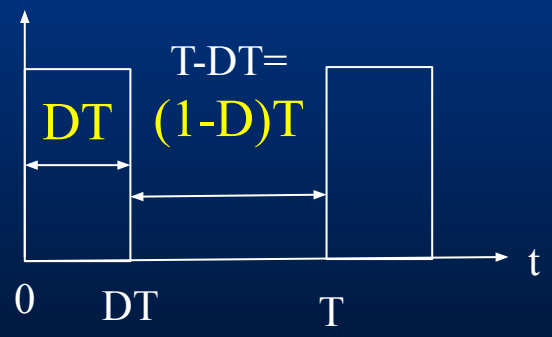



- 입력(Vs)보다 출력(Vo)이 D=0.5이상이면 커짐

# 6.3 Buck-Boost Converter: 전압




- Switch On

$$v_L = V_S = L\frac{di_L}{dt} = L\frac{\Delta i_L}{\Delta t} = L\frac{\Delta i_L}{DT}$$

$$\therefore \Delta i_L = \frac{1}{L}V_S DT$$

- Switch Off

$$v_L = -V_0 = L\frac{di_L}{dt} = L\frac{\Delta i_L}{\Delta t} = L\frac{\Delta i_L}{(1-D)T}$$

$$\therefore \Delta i_L = -\frac{1}{L}V_0(1-D)T$$

- 충전전류+방전전류=0

$$\cancel{\frac{1}{L}}V_S D\cancel{T} \times \frac{L}{T} - \cancel{\frac{1}{L}}V_0(1-D)\cancel{T} \times \frac{L}{T} = 0 \times \frac{L}{T}$$

$$V_S D - V_0 + V_0 D = 0$$

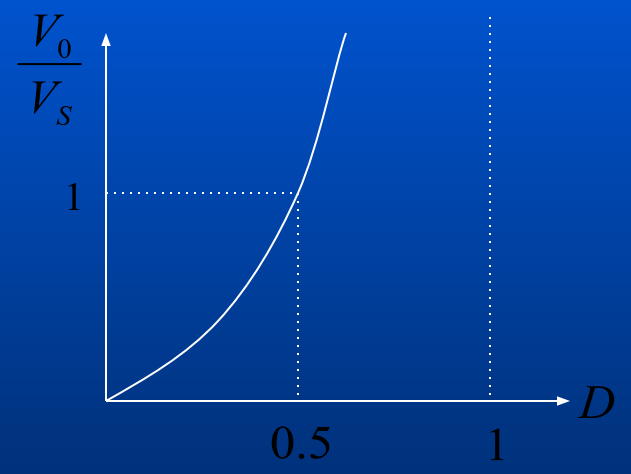
$$\therefore V_0 = \frac{V_S D}{(1-D)}$$

# 6.6 Buck-Boost Converter: 출력

- 충전전류+방전전류=0

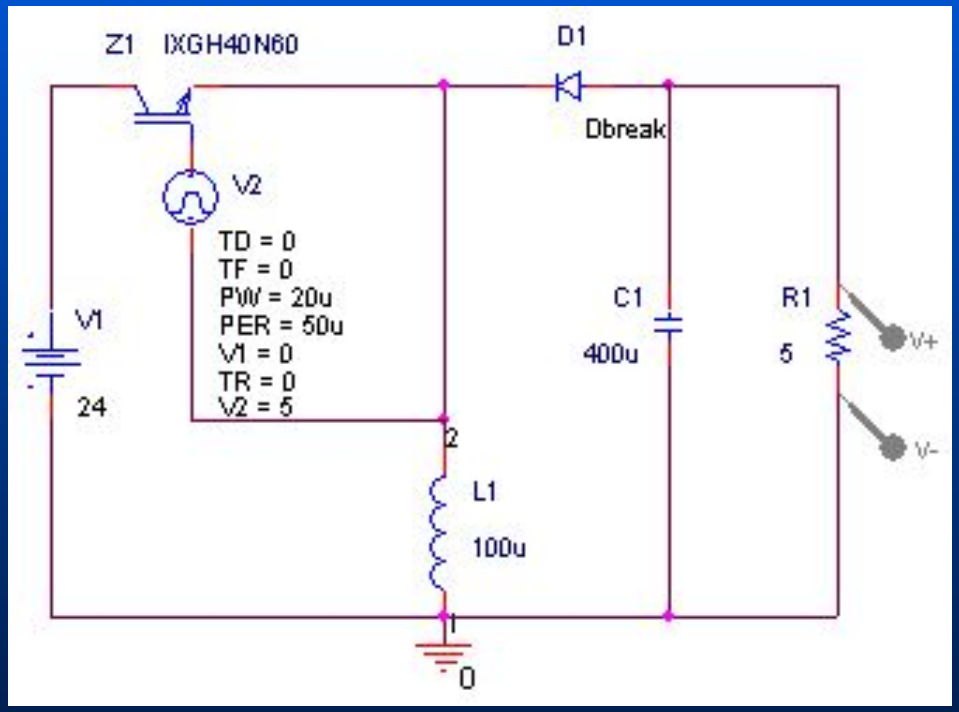
$$D \le 1, \quad V_0 = V_S \frac{D}{(1-D)}, \qquad \therefore \frac{V_0}{V_S} = \frac{D}{(1-D)}$$

| D | 출력/입력 | $(V_0/V_S)$ |
|---|---|---|
| 0 | 0/(1-0) | 0 |
| 0.2 | 0.2/(1-0.2) | 0.25 |
| 0.5 | 0.5/(1-0.5) | 1.0 |
| 0.8 | 0.8/(1-0.8) | 4.0 |
| 1.0 | 1/(1-1) | ∞ |



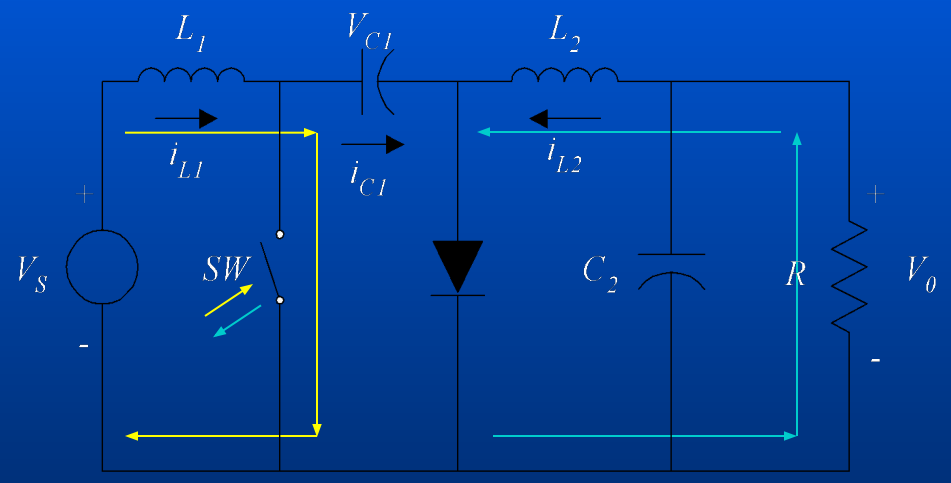

- 강압 및 승압 (Step-up & down)
- 입력 대비 출력이 낮거나 높음

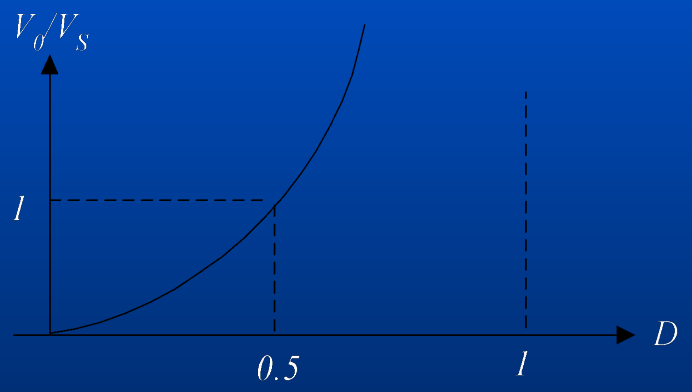
# 예제 6-4 벅-부스트 변환기(Pspice 해)




- *Parameter*
  - ✔ $V_S$=24 V
  - ✔ R=5 ohm
  - ✔ L=100 μH
  - ✔ C=400 μF
  - ✔ f=25 kHz(T=40 μS)
  - ✔ D=0.6 (D=24 μS)
  - ✔ Step
    =0 0.5 ms
- *Find*
  - ✔ 출력전압
  - ✔ 전류 최대, 최소값

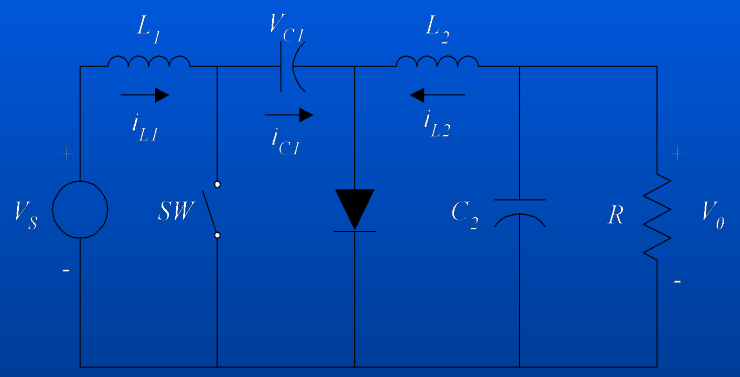
# 6.6 Cuk Converter




- 입력($E_1$) VS 출력($E_2$) 이 적거나 커짐

# 6.7 Cuk Converter: 전압




- Switch On

$$V_{C1} = V_S - V_0$$

$$(i_{C1})_{closed} = -I_{L2}$$

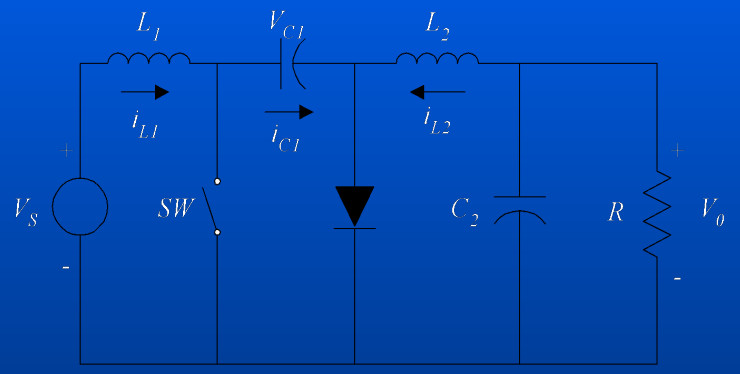
- Switch Off

$$(i_{C1})_{open} = I_{L1}$$

- 충전전류+방전전류=0

$$-i_{L2}DT + i_{L1}(1-D)T = 0, \quad \frac{i_{L1}}{i_{L2}} = \frac{D}{(1-D)}$$

# 6.7 Cuk Converter: 출력



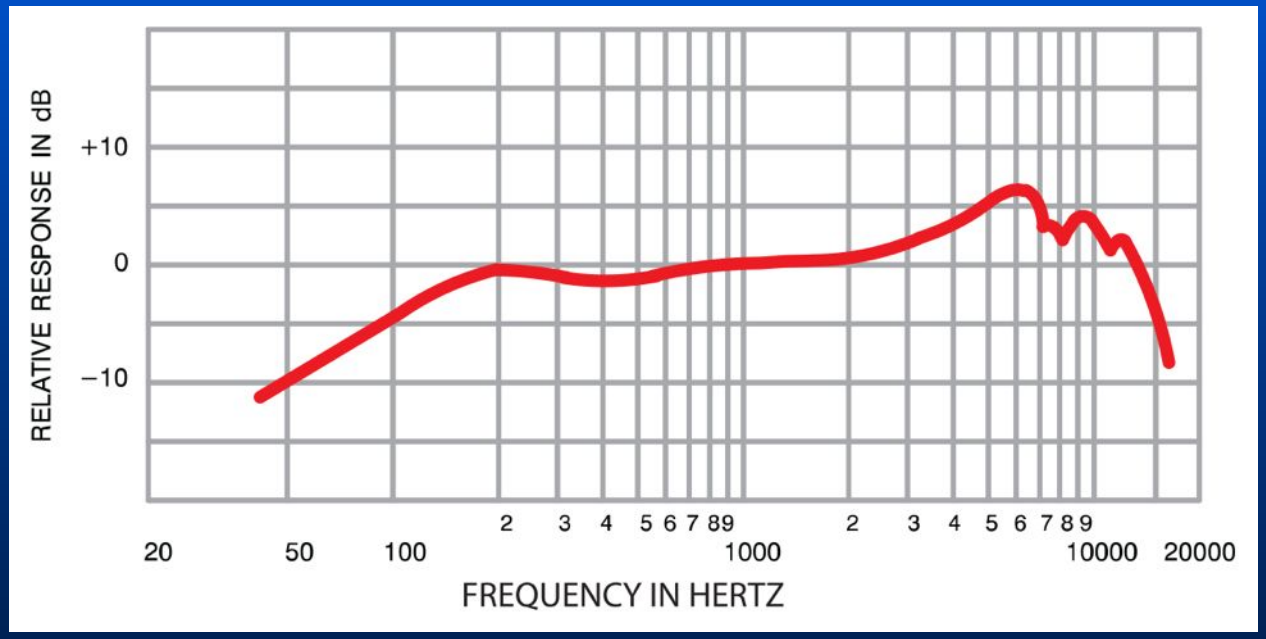

- 공급전력=소모전력

$$P_S = P_0$$

$$V_S i_{L1} = -V_0 i_{L2}, \quad \frac{i_{L1}}{i_{L2}} = \frac{-V_0}{V_S}$$

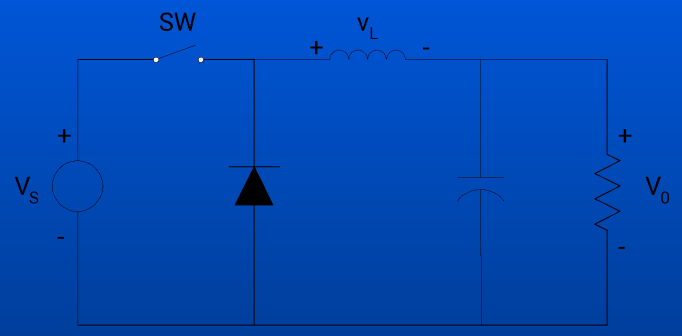
$$\therefore \frac{V_0}{V_S} = -\frac{D}{(1-D)}, \quad \left( \because \frac{i_{L1}}{i_{L2}} = \frac{D}{(1-D)} \right)$$

# Switching Frequency

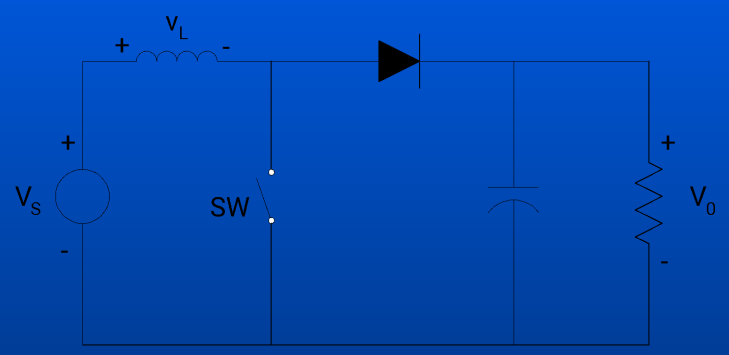



- Audio frequency(가청주파수)을 벗어나는 20kHz 이상의 초음파(ultra sonic) 영역에서 스위칭하여 소음을 피할 수 있음

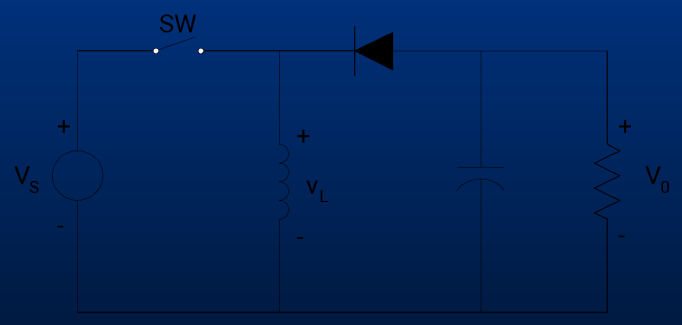
# DC-DC Converter 종합



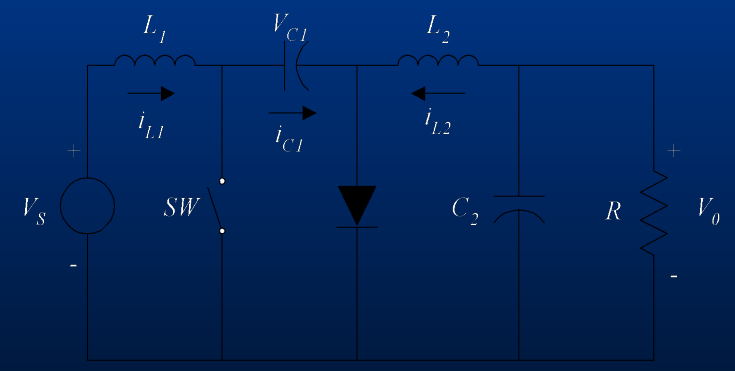

- Buck: 강압




- Boost : 승압




- Buck-boost: 승압/강압




- Cuk: 승압/강압